Runi: Locality-Guided GPU Reoptimization for Batched Dynamic Capacitated Vehicle Routing

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Abstract

Reoptimizing vehicle routes after customer cancellations and new requests provides an opportunity to reuse high-quality existing tours instead of solving the full capacitated vehicle routing problem from scratch. We present Runi, a CUDA-based batch-repair method that combines centroid-guided candidate caching, exclusive route-worker warps, ready-route dispatch, a multi-block scheduler, and spatially sparse local-search refinement. The method separates fast feasibility repair from solution-quality improvement through route-tail exchanges, customer relocations, SWAP*, and intra-route 2-opt. In the recorded experiments, Runi completes valid balanced update batches of up to 12,800 events on a 500,000-customer instance. A refinement sweep identifies 15 passes as a practical quality–latency compromise for the tested workloads. Five-run 80/20 reconstruction experiments yield mean gaps of 4.24–7.70% relative to the current CVRPLib best-known solution values, with run-to-run variation reported explicitly. We also compare Runi with Senay Patel's dynamic VND implementation [2] and the static seqMDS approach [1], while distinguishing comparisons with different hardware and optimization scopes from direct dynamic-update evidence.

A reference page at the end of this report collects key technical terms, notation, hyperparameters, and benchmark instance details for quick reference.

1. Introduction

The Capacitated Vehicle Routing Problem (CVRP) asks for a set of depot-to-depot routes that serves every customer exactly once while respecting the capacity of each vehicle and minimizing total travel cost. It is a classical NP-hard optimization problem, and a large body of work has developed constructive heuristics, metaheuristics, and local-search methods for producing high-quality static routes. In a static formulation, the customer set is assumed to be known before optimization begins. Operational routing systems, however, must often revise an existing plan after it has already been constructed.

1.1 The dynamic routing gap

In delivery networks, ride-hailing services, and other customer-facing logistics systems, the active request set can change during operation. A new order may need to be served, a customer may cancel, or a customer's demand may be revised. These changes turn a static CVRP instance into a dynamic one: the solver must update a feasible incumbent solution in response to a sequence or batch of events, rather than solve only one fixed customer set [7]. A natural but expensive response is to discard the incumbent and solve the modified instance again. That approach repeats work on customers and route segments that have not changed. At the other extreme, a purely greedy insertion can be fast but may leave poor assignments that require broader route changes.

The objective of this work is to balance these two concerns. Runi treats an update as a planning snapshot: it applies all deletions, places the new or displaced requests, and then refines the resulting solution. The current route plan is retained as a useful starting point, but the refinement stage is allowed to change route membership and order when a better feasible solution is found. This is a batched reoptimization model; it does not model vehicle movement or immutable route prefixes that have already been executed.

1.2 Computational challenges at scale

The challenge is not only to find good changes, but to find them without repeatedly examining the entire routing structure. If every new request scans all route members, or if refinement considers every pair of routes, the work grows quickly with the instance. A dense all-pairs distance table can also become prohibitive: storing one double for every ordered pair requires about 74.5 GiB at 100,000 nodes and about 1.82 TiB at 500,000 nodes. Such a table is useful in some static local-search implementations, but its quadratic storage is an unattractive basis for very large dynamic workloads.

GPU execution adds a second constraint. Many requests can be processed in parallel, but two workers must not concurrently rewrite the same route. Route updates also change loads, centroids, and the set of promising refinement partners. A scalable design therefore needs both a way to limit the candidate routes considered for

each request and a synchronization scheme that permits independent routes to be processed concurrently. RUNI addresses these requirements with centroid-guided candidate caching, a multi-block scheduler, a ready-route queue, and route-exclusive worker warps.

1.3 Contributions

This work makes three contributions. First, it presents a batched DELETE–ADD–refine pipeline in which new and ejected customers share a global frontier and are assigned to capacity-feasible routes. Second, it develops a GPU scheduling design that scans route centroids across multiple blocks, dispatches only routes with pending work, and gives each worker warp exclusive ownership of the route it changes. Third, it combines sparse inter-route refinement (including depot-edge extensions to Moves 8/9) with relocation, SWAP*, and intra-route 2-opt, and evaluates the quality–latency trade-off across large instances. The evaluation includes five independent runs for each 80/20 reconstruction instance, so mean cost, sample standard deviation, and the observed BKS-gap range can be reported rather than relying on a single run.

2. Background and Related Work

CVRP research provides several complementary foundations for dynamic route optimization: population-based static search, neighbourhood-based local improvement, and GPU parallelization of large route computations. This section summarizes the approaches most closely connected to RUNI and clarifies the distinction between a static restart and an update to an existing solution.

2.1 Static CVRP optimization: HGS and granular local search

Hybrid Genetic Search (HGS) is a prominent metaheuristic for static CVRP instances. It combines population-based exploration with an intensive local-search procedure that improves individual solutions [3]. The population provides diversity, while local-search moves improve route order and the assignment of customers to vehicles. Some HGS formulations also use penalty terms for capacity or duration violations during search, permitting exploratory solutions before feasibility is restored. This differs from RUNI's repair path, which commits only capacity-feasible route changes. HGS-style methods commonly use granular neighbourhoods: rather than evaluating every possible customer pair, each customer considers a limited set of nearby candidates. This idea reduces the number of local-search moves that need to be examined and is closely related to granular search strategies in vehicle routing [8].

The local-search neighbourhoods relevant to this report include customer relocation, exchanges between routes, intra-route 2-opt, and inter-route tail exchanges (2-opt*). SWAP* extends a simple one-for-one exchange by allowing the two exchanged customers to be reinserted at their best positions in the opposite routes [3]. These operators provide useful building blocks for improving a feasible CVRP solution. RUNI does not implement the HGS genetic algorithm or its full search framework; it uses selected route-improvement neighbourhoods within a GPU-oriented dynamic update pipeline. In particular, its refinement stages use Moves 8/9, Move 1, SWAP*, and Move 7 (2-opt), with capacity-feasible changes checked before they are committed.

2.2 Variable Neighbourhood Descent and Senay Patel's dynamic solver

Variable Neighbourhood Descent (VND) systematically explores a sequence of neighbourhoods. When an improving move is accepted, the search can return to an earlier neighbourhood because the modified route structure may expose new opportunities; termination occurs when a full scan finds no further improvement [6]. This makes VND a natural way to recover solution quality after an insertion or deletion, but its cost depends on the number of candidate moves examined and the number of accepted improvements.

Senay Patel's Dynamic-CVRP solver is a directly relevant dynamic baseline. It supports customer insertion and deletion and applies VND-style local search to improve the affected solution [2]. Its implementation provides a useful reference for the move terminology and for comparing dynamic repair against a CPU-based neighbourhood-search approach. In this report, Senay's solver is treated as a baseline, not as part of RUNI's implementation. RUNI adopts selected improvement operators but changes the execution model: requests are processed in batches, route candidates are ranked using centroids, and GPU worker warps mutate separate routes under explicit ownership. The refinement is also bounded by a pass budget and stops early when a complete pass commits no improvement.

2.3 seqMDS and GPU-parallel vehicle routing

The seqMDS method of Muniasamy et al. studies parallelization of the static vehicle routing problem [1]. Its construction uses an MST-guided traversal: randomized depth-first traversals produce customer visit orders, which are divided into capacity-feasible routes before repeated improvement. This makes seqMDS a relevant static GPU-oriented point of comparison. However, it begins from a fixed instance and constructs or improves a full solution; it is not an event-driven mechanism for modifying an incumbent after a small batch of customer changes. In this report, seqMDS is therefore used as a static-restart comparison, and its timing is not interpreted as a directly matched dynamic-update speedup.

More broadly, GPU work on routing has explored parallelizing expensive parts of search, including neighbourhood evaluation and local-search computations [9, 1]. Such work demonstrates the value of exposing parallelism in route

optimization. Dynamic updates introduce an additional requirement: parallel workers must coordinate access to mutable route state, because two candidate moves that touch the same route cannot safely commit independently. This conflict-management problem is central to RUNI's ready-route dispatch and conflict-free refinement rounds.

2.4 Position of Runi

The methods above address different parts of the problem. HGS and related local-search frameworks focus on solution quality for static instances; seqMDS explores GPU parallelism for static routing; and Senay Patel's solver provides a dynamic insertion/deletion baseline with VND-based improvement. RUNI combines the dynamic and GPU perspectives: it retains the incumbent, handles a batch of deletions and additions through a shared request frontier, uses centroid-ranked routes to limit scheduling work, and assigns each active route to one worker warp at a time. A sparse refinement phase then improves the repaired solution. The goal is not to replace the full range of static CVRP metaheuristics, but to provide a practical route-repair and refinement path for large batched updates.

3. Problem Statement

Consider a single depot 0 and a set of active customers V_t at update epoch t. Customer i has demand $q_i > 0$ and Euclidean location $\mathbf{p}_i = (x_i, y_i)$, and all vehicles have capacity Q. A solution $\mathcal{R}_t$ is a set of ordered routes $r = (0, v_1, \ldots, v_{L_r}, 0)$ such that every active customer appears exactly once, no inactive customer is served, and

$$\sum_{i \in r} q_i \le Q \qquad \forall r \in \mathcal{R}_t. \tag{1}$$

We minimize the sum of travel distances,

$$\begin{aligned} F(\mathcal{R}_t) &= \sum_{r \in \mathcal{R}_t} \sum_{(u,v) \in E(r)} d(u,v), \\ d(u,v) &= \operatorname{round}(\|\mathbf{p}_u - \mathbf{p}_v\|_2)\,. \end{aligned} \tag{2}$$

with rounding applied separately to each edge.

An update consists of a deletion set D_t and an addition set A_t, taking V_t to $V_{t+1} = (V_t \setminus D_t) \cup A_t$. The initial routes are given. RUNI processes the entire transaction in the order *all deletions, all additions, then refinement*; this semantics does not depend on the order in which records were listed in the input. Unmodified portions of routes may be changed during refinement, so this is an *unrestricted planning-snapshot* model.

The benchmark is an unrestricted reoptimization snapshot: completed travel, vehicle motion, and committed route prefixes are not modeled, and the objective is evaluated solely on the final active customer set.

4. Design Motivation: From Static seqMDS to Dynamic Updates

With seqMDS introduced in Section 2, we next describe the design question that led to RUNI: whether a dynamic solver should maintain the static construction structure, or preserve the current routes and spend its work on the changed part of the solution. The following profiling observation motivated the latter choice.

4.1 Why not maintain the minimum spanning tree?

The starting point for this investigation was the static *seqMDS* algorithm [1]. Its construction first obtains a minimum spanning tree (MST), explores randomized depth-first traversals of that tree, splits the resulting visit order into capacity-feasible routes, and repeatedly searches for improvements. This is effective for a new, static instance, but it is not automatically a good strategy when only a small fraction of customers change.

An initial direction was to maintain the MST incrementally as customers were inserted or deleted. The attraction is clear: if the tree could be repaired cheaply, perhaps the updated routes could be constructed without restarting the static algorithm. Profiling shifted our focus. In the examined static execution, approximately **99% of time** was spent in the repeated optimization *superloop*, rather than the one-time MST construction. A faster MST update would therefore accelerate only a small portion of execution. The more promising intervention was to avoid repeating the whole route-construction and improvement process after each batch.

This observation led to an architectural change rather than a more elaborate dynamic-tree data structure: retain the current solution, identify a small set of promising modifications, and invoke the expensive improvement neighborhoods selectively. The next sections explain how RUNI turns that principle into a GPU pipeline.

5. Runi: Locality-Guided GPU Repair

The central idea of RUNI is to preserve useful incumbent routes while repairing only what a batch of customer changes makes necessary. This section describes how deletions, centroid-guided scheduling, and route-exclusive GPU workers cooperate to restore a feasible solution before refinement.

5.1 From cancellations to unplaced requests

Deletions are simpler than additions. RUNI marks all cancelled customers, compacts every affected route, reconnects the surrounding edges, and refreshes route load and geometric metadata. Surviving customer order is preserved during this fast repair step. New customers form a device-resident *frontier* of requests that still need a route. During *ejection*, a worker creates room for a newly inserted customer by removing an existing customer from a route; that displaced customer returns to the same frontier to be assigned elsewhere.

This shared representation is important: instead of implementing separate procedures for every failure mode, every unsettled request—brand-new or ejected—re-enters the same placement loop until it is inserted or declared infeasible. A scheduler decides *which route should be tried*; a worker then decides *how that route should be changed*. Figure 1 shows the multi-block scheduler, ready-route queue, and per-route work queues.

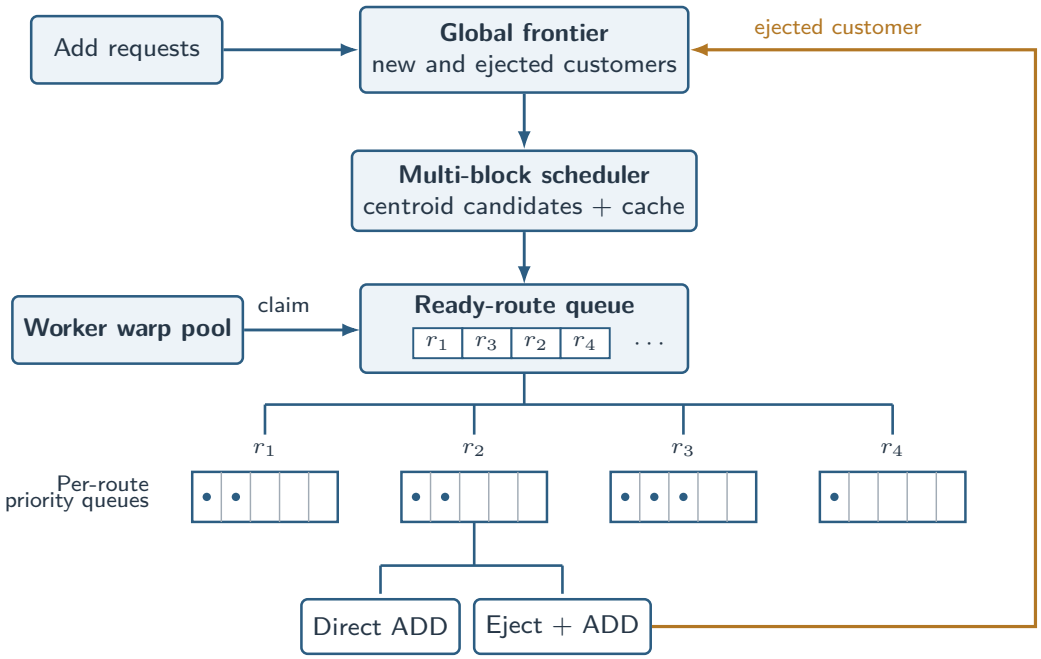
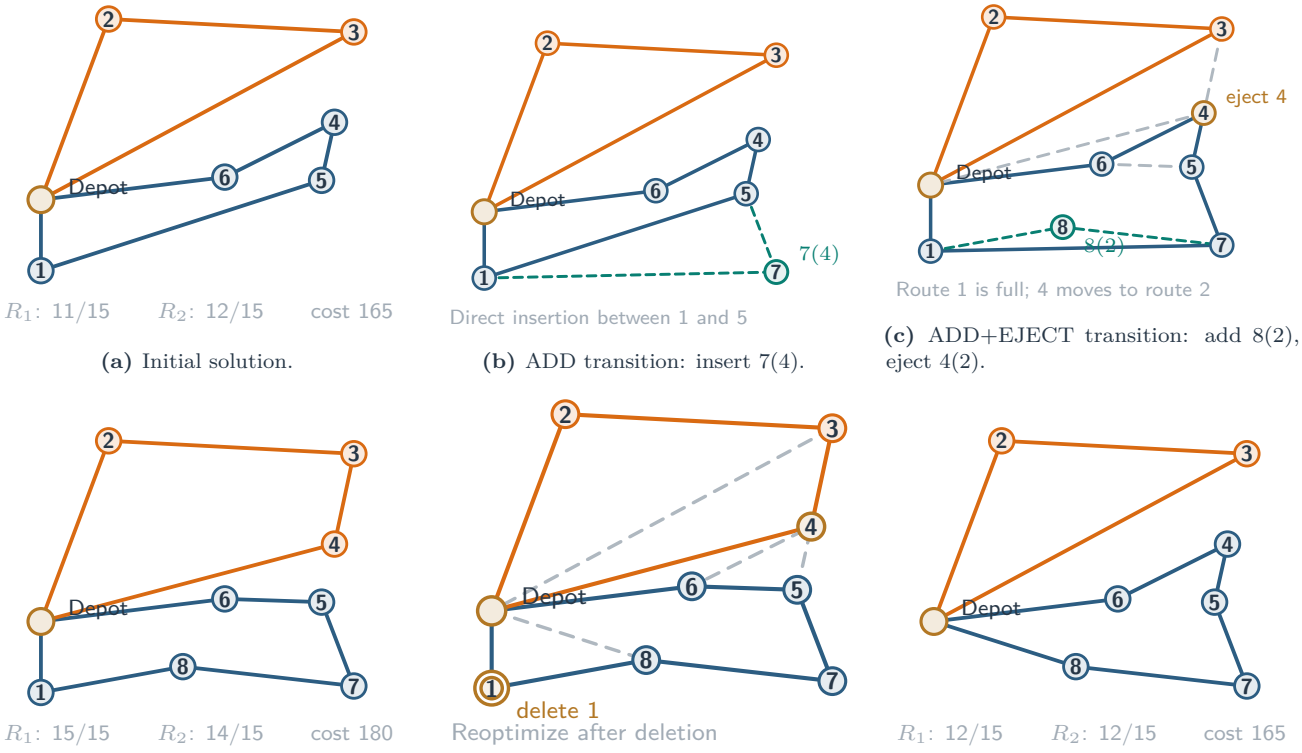


Figure 1: Runi's multi-block ADD/repair pipeline. Ejected requests return to the frontier; repaired routes are then passed to refinement.

Algorithm 1 Runi: one batched dynamic-CVRP update

Require: Feasible incumbent routes $\mathcal{R}$, deletions D, additions A, pass budget P

DELETE: Remove D and refresh affected routes' loads and centroids.
Prepare ADD: Put all new requests A in the global frontier.
while frontier requests, route-queue tasks, or active workers remain **do**
    The multi-block scheduler assigns requests to promising routes using centroids.
    Route-worker warps claim routes exclusively and attempt direct insertion.
    If insertion needs ejection, return each displaced request to the frontier.
end while
Refine once after repair:
for $p = 1, \ldots, P$ **do**
    Apply Moves 8/9, then Move 1/SWAP*, then intra-route 2-opt.
    if none of the three stages committed an improving move **then**
        break ▷ Stop early when the current pass makes no progress
    end if
end for
Validate route feasibility and return the updated solution.



(a) Initial solution. **(b)** ADD transition: insert 7(4). **(c)** ADD+EJECT transition: add 8(2), eject 4(2).

(d) After ADD+EJECT. **(e)** DELETE transition: remove 1(5). **(f)** Final repaired solution.

Figure 2: Running example ($Q = 15$): direct ADD of 7(4); ADD 8(2) with ejection of 4(2); DELETE 1(5) and reoptimization. The final cost returns from 180 to 165. Dashed edges indicate proposed changes.

The running trace in Fig. 2 illustrates the three route-editing cases used throughout the implementation. The initial solution has route loads 11/15 and 12/15. Adding customer 7(4) is a direct insertion, yielding route 1 load 15/15 and cost 175. Adding customer 8(2) cannot be absorbed by that same route without violating capacity, so route 1 ejects customer 4(2) and places customer 8(2) instead; customer 4 is then reassigned to route 2, producing route loads 15/15 and 14/15 at cost 180. Finally, deleting customer 1(5) triggers another reoptimization in which customer 4 moves back to route 1, restoring a total cost of 165. The example is illustrative rather than a transcript of a single GPU warp, but it matches the semantics implemented by the repair stage.

5.2 Looking near the customer, not everywhere repeatedly

For route r with L_r customers, RUNI maintains a depot-inclusive centroid

$$\mathbf{c}_r = \frac{\mathbf{p}_0 + \sum_{i \in r} \mathbf{p}_i}{L_r + 1}. \tag{3}$$

A centroid gives one constant-size spatial summary per route, so ranking R routes requires only $O(R)$ distance checks. Using the nearest customer vertex of every route instead would require examining many route members (or maintaining an additional spatial index) every time a new request is assessed. The centroid is therefore a fast locality heuristic, not a guarantee of the route with the cheapest feasible insertion. The centroid can also be refreshed incrementally after a small route edit, reducing the need to revisit every customer during candidate ranking.

Figure 3 gives a concrete example. Let the depot be $(50, 50)$ with vehicle capacity $Q = 15$. Route 1 contains customers $1 = (40, 60)$ and $2 = (55, 65)$ with demands 6 and 7, for load 13/15. Route 2 contains customers $3 = (49, 57)$ and $4 = (53, 49)$ with demands 4 and 7, for load 11/15. Now add $x = (48, 61)$ with demand 1. The new request is geometrically close to customer 3:

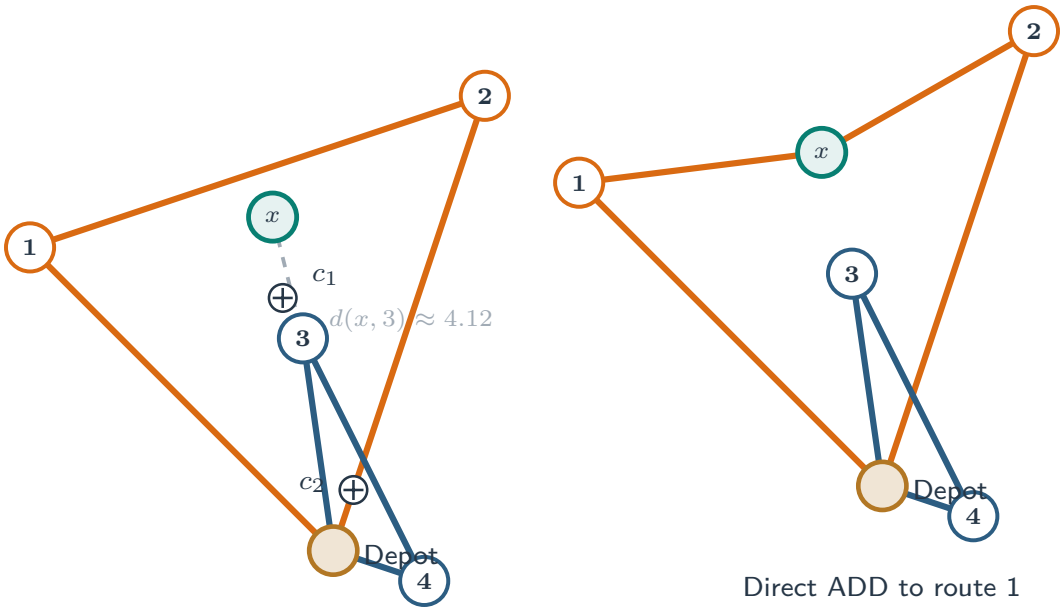
$$d(x, 3) = \sqrt{(48 - 49)^2 + (61 - 57)^2} = \sqrt{17} \approx 4.12. \tag{4}$$

A nearest-vertex heuristic would therefore be tempted to try route 2 first. The depot-inclusive centroids, however, are

$$\mathbf{c}_1 = \left(\frac{50 + 40 + 55}{3}, \frac{50 + 60 + 65}{3}\right) = (48.33, 58.33), \qquad \mathbf{c}_2 = \left(\frac{50 + 49 + 53}{3}, \frac{50 + 57 + 49}{3}\right) = (50.67, 52.00), \tag{5}$$

with distances

$$d(x, \mathbf{c}_1) \approx 2.69, \qquad d(x, \mathbf{c}_2) \approx 9.39. \tag{6}$$



(a) Before insertion: $x = (48, 61)$, $q_x = 1$. **(b)** After insertion into route 1.

Figure 3: Centroid-guided route selection example. Although x is closest to customer 3, the depot-inclusive centroid ranking prefers route 1, which also has the lower rounded insertion cost.

Thus the centroid shortlist correctly favors route 1. That choice is also better under the rounded insertion metric. On edge $(1, 2)$,

$$\Delta_{\text{ins}}(1, x, 2) = d(1, x) + d(x, 2) - d(1, 2) = 8 + 8 - 16 = 0, \tag{7}$$

whereas the best feasible insertion into route 2 has cost increase 8. This example shows why centroid guidance is attractive: it is cheaper than checking every vertex of every route, yet it can still rank routes according to the larger spatial structure rather than a single nearby customer. The heuristic is not exact for all placements, which is why the later refinement stage remains necessary.

5.3 The ready-route queue and the worker warps

A selected customer is placed in the bounded priority queue of its chosen route. The route's identifier is then published to a ready-route queue, so idle workers can discover exactly which route has work. A worker warp claims one route exclusively, removes its highest-priority pending request, performs the modification, and relinquishes ownership. The ownership protocol prevents two workers from concurrently rewriting the same route; a lightweight reservation also protects the scheduler's queue insertion.

Within a route, 32 CUDA lanes evaluate insertion positions cooperatively. Inserting x between u and v changes distance by

$$\Delta_{\text{ins}}(u, x, v) = d(u, x) + d(x, v) - d(u, v). \tag{8}$$

The worker selects a feasible low-cost insertion using warp-level reduction. If capacity is insufficient, it can instead eject an existing customer y while checking $\sum_{i \in r} q_i - q_y + q_x \le Q$; the displaced request returns to the frontier. When eligible routes have been exhausted, a feasible singleton $(0, x, 0)$ can be created. The repair stage terminates only when no frontier requests, route-queue entries, or active route workers remain. The resulting solution is feasible before refinement begins. In the running example, the scheduler enqueues x or y for a route, but only the owning worker warp is allowed to modify that route; the displaced d is handled as another frontier request.

5.4 Multi-block scheduling without losing warp parallelism

For a large route set, a single scheduler warp leaves too much sequential scanning. RUNI assigns approximately ρ candidate routes to each scheduler thread; with R possible routes and 32 lanes per warp, the nominal scheduler width is

$$W_s = \left\lceil \frac{R}{32\rho} \right\rceil, \qquad \rho = 8 \text{ routes/thread}. \tag{9}$$

Unlike a one-block scheduler, the route scan is partitioned across cooperating blocks. Block 0 retains coordination and combines per-block candidate summaries, while the other blocks evaluate disjoint route subsets for the same frontier customer. This preserves one-customer-at-a-time decisions and limits changes to queue correctness, while extending scanner parallelism beyond the 32-warp limit of one CUDA block.

A second constraint is less obvious. A worker warp temporarily stores a route sequence during mutation. If B warps share a block and the reserved route capacity is S integer entries, its dynamic shared-memory request is $4BS$ bytes. As S grows, keeping the same number of warps per block can exceed the available shared memory

Algorithm 2 Exclusive route-worker warp (one claimed route at a time)

while the ready-route queue contains work **do**
  Claim a ready route r exclusively and pop its next customer x.
  Evaluate insertion positions in r cooperatively across the warp.
  if x fits and a feasible direct insertion is found **then**
    Insert x at its lowest-cost feasible position.
  else if a feasible eject-and-insert change is found **then**
    Insert x; return the displaced customer to the frontier.
  else
    Return x to the scheduler for another candidate route.
  end if
  Refresh route state, release ownership, and republish pending work.
end while

even though the total number of desired scheduler warps has not changed. RUNI therefore varies warp packing independently from W_s. For large route strides, when shared-memory packing would leave only one warp per block, worker scratch is moved to preallocated global memory, allowing the scheduler and worker grid to use wider blocks. The scheduler's cross-block barriers are then shared among fewer blocks, rather than among many single-warp blocks. A residency-aware launch configuration keeps the cooperative scheduling grid resident to prevent barrier deadlock.

Algorithm 3 Multi-block centroid-guided scheduler

while an unplaced customer x remains in the frontier **do**
  Reuse valid cached route candidates; refresh changed route centroids.
  Let scheduler blocks scan disjoint groups of route centroids.
  Combine the blocks' nearest-route proposals into a shortlist.
  Choose an untried candidate, accounting for capacity and route availability.
  if a route is available **then**
    Enqueue x in that route's priority queue; publish its route ID.
  else
    Try the fallback candidates or open a feasible singleton route.
  end if
end while

6. Sparse GPU Refinement

Feasibility alone does not ensure short routes: locally convenient insertions may leave costly assignments or detours. After repair, RUNI therefore applies a bounded sequence of nearby-route exchanges, customer moves, and intra-route improvements to reduce travel cost without restarting the optimization from scratch.

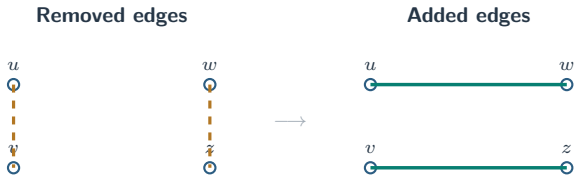
6.1 Why repair alone is insufficient

The repair phase has a narrow job: make the updated solution feasible quickly. A new customer may be inserted into a locally attractive route without accounting for broader improvements involving neighboring routes. Stopping here would systematically trade away solution quality. RUNI therefore adds a distinct refinement stage, called once per batch, in which a bounded number of internal passes reshapes the routes after all requests have been placed.

An exhaustive search over all unordered route pairs would examine $R(R-1)/2$ possibilities. Instead, each live route nominates its K_r nearest neighboring route centroids; after deduplication, the candidate set contains at most $K_r R$ directed nominations and usually fewer unique undirected pairs. These pairs are greedily separated into conflict-free matching rounds: no route appears twice in the same round. A warp can then evaluate and commit a pairwise improvement without another warp simultaneously changing either route. Each round is completed before the next begins, so later evaluations see the new route state.

Two-edge exchange in the running example. After all customers (including displaced d) have been placed, refinement may examine an incumbent tour segment $\cdots, u, v, \ldots, w, z, \cdots$. A 2-opt move removes two edges (u, v) and (w, z) and inserts (u, w) and (v, z), reversing the intervening segment. Its distance change is

$$\Delta_{\text{2opt}} = d(u, w) + d(v, z) - d(u, v) - d(w, z). \tag{10}$$



Only a negative change is committed. This two-edge replacement is different from an ADD/DELETE event: it improves visit order without changing the active customer set or route load. Nearby-route matching similarly ensures that concurrent inter-route moves do not mutate the same route.

6.2 The three refinement neighborhoods

The refinement is VND-inspired: it applies a fixed sequence of neighbourhood groups and repeats the sequence while passes continue to improve the solution. Senay Patel's Dynamic-CVRP implementation is the direct CPU baseline for this style of post-event repair [2]; the general VND principle is described by Hansen and Mladenović [6], while SWAP* is the specialized customer-exchange neighbourhood introduced for CVRP search [3]. RUNI does not reproduce the baseline's complete search procedure or the full HGS move set. Instead, it selects three groups suited to its GPU execution model: route-tail exchange (Moves 8/9), inter-route relocation and SWAP*, and intra-route 2-opt (Move 7). The numbered names refer to the implementation's move conventions.

Stage I—Move 8/9, route-tail exchange [2]. Pairs of nearby routes exchange suffixes, considering the relevant orientation of each exchanged tail. This can move entire groups of customers between vehicles and repair a poor division created by sequential insertion. Candidate moves must preserve capacities and decrease total route cost.

Algorithm 4 Refinement stage I: nearby-route tail exchange (Moves 8/9)

```
Form nearby, non-overlapping pairs of routes.
for each pair (r, s) in the current parallel round do
    Examine feasible tail exchanges, including cuts at depot edges.
    Choose the largest distance reduction that preserves vehicle capacity.
    if the change decreases total rounded route cost then
        Apply the exchange and refresh both routes.
    end if
end for
```

Depot-edge extension for Moves 8/9. The original customer-edge search restricts cut endpoints to customer vertices. We extend it to include depot-adjacent edges, allowing a tail-exchange move to *join two routes at the depot* whenever their combined load permits. For example, with $Q = 15$, let $R_1 = (0, 1, 3, 0)$ have load $5 + 3 = 8$ and $R_2 = (0, 4, 2, 0)$ have load $5 + 2 = 7$, while a third route $(0, 6, 5, 0)$ has load $7 + 7 = 14$ and remains unchanged. Cutting $B = (3, 0)$ and $A = (2, 0)$ and introducing $C = (3, 2)$ produces the feasible merged tour $(0, 1, 3, 2, 4, 0)$ of load 15/15; the second route becomes empty.

$$\Delta_{\text{merge}} = d(3, 2) - d(3, 0) - d(2, 0) = C - A - B. \tag{11}$$

In unrounded Euclidean geometry, triangle inequality gives $C \leq A + B$ (strict for non-collinear points). Since this study rounds distances edge by edge, the implementation should still accept the move only when the *rounded* $\Delta_{\text{merge}} < 0$. Including depot cuts can replace a sequence of customer relocations with one inter-route move, shortening the improvement process in favorable cases; it does not guarantee improvement for every pair.

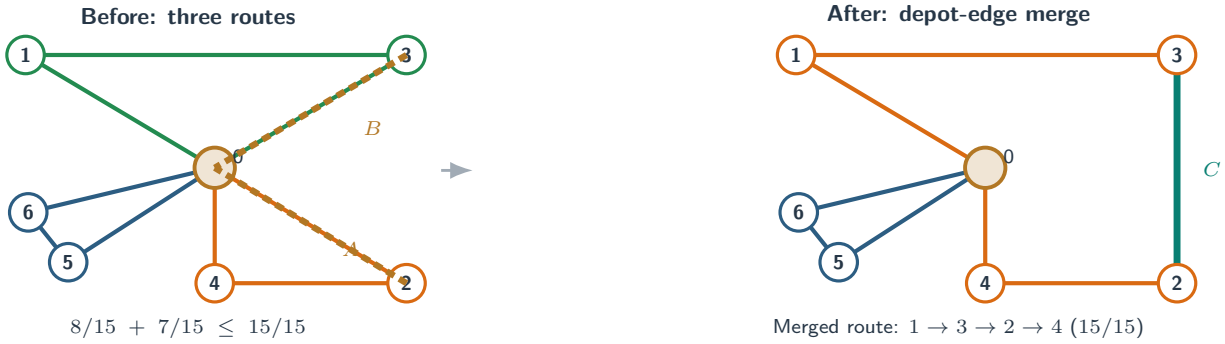


Figure 4: Depot-edge Move 8/9 extension. Removing the depot edges $A = (0, 2)$ and $B = (3, 0)$ and adding $C = (3, 2)$ merges two capacity-compatible routes. The third route is unchanged.

Stage II—Move 1 and SWAP* [2, 3]. After Stage I, the centroids and the nearby-pair nominations are rebuilt because route membership has changed. Move 1 relocates a customer between routes. SWAP* exchanges customers between two routes while permitting their best reinsertion positions rather than a fixed position-for-position swap. These moves refine individual customer assignments after the coarser tail exchanges.

Algorithm 5 Refinement stage II: relocation and SWAP*

```
Rebuild nearby-route pairs after the tail-exchange stage.
for each non-overlapping pair of routes (r, s) do
    Evaluate moving one customer to the other route (Move 1).
    Evaluate exchanging two customers with best reinsertion (SWAP*).
    Keep a capacity-feasible move only if it reduces route cost.
end for
```

Stage III—Move 7, intra-route 2-opt [2]. Each live route is assigned a warp to consider improvements to its internal visiting order. Reversing an appropriate segment can remove an inefficient detour without changing vehicle capacity. Since routes are independent here, many 2-opt evaluations can proceed simultaneously.

Algorithm 6 Refinement stage III: intra-route 2-opt (Move 7)

```
for each active route r in parallel do
    Search for two edges that can be reconnected more cheaply.
    if reversing the segment reduces rounded distance then
        Apply the reversal without changing the route's customer set.
    end if
end for
```

A refinement budget P means *at most* P complete passes through these stages. After each pass, the implementation sums the committed changes from Moves 8/9, Move 1/SWAP*, and 2-opt. If the total is zero, refinement stops immediately; otherwise, it continues until the budget is reached. In the results we distinguish RUNI1 ($P = 1$), used to examine low-latency dynamic updates, from RUNI15 ($P = 15$), selected for stronger final-solution quality. The passes are part of the same single refinement invocation after a transaction, not one refinement invocation per ADD or DELETE event.

7. Experimental Design

We evaluate Runi on route sets of increasing size, separating feasibility, solution quality, and update latency rather than treating them as interchangeable metrics.

7.1 Hardware, software, and timing scope

The GPU benchmark record identifies an NVIDIA GeForce RTX 5090 (Blackwell, CUDA compute capability 12.0). The project is CUDA/C++ with an SM 120 build target; CUDA 13.2 tooling is present on the profiling host. The supplementary profiles were collected using Nsight Compute and Nsight Systems, with the Systems report identifying version 2025.6.3. Individual observations are claimed to be multi-run averages.

7.2 Benchmarks and measurements

The experiments use CVRPLIB-style instances [5] with depot-to-depot tours, positive customer demands and per-edge rounded Euclidean distances. They test three distinct forms of difficulty: 1,000 customers with relatively few routes (X-n1001-k43), 10,000 customers with many short routes (XL-n10001-k1570), and 30,000 customers with relatively long routes (Flanders2). A larger 100,000-customer instance tests mixed updates, and a 500,000-customer instance tests the ability to accommodate growing event batches. Exact names, capacities and dimensions are collected in the final reference page.

A balanced mixed batch contains m deletions and m additions, resulting in $2m$ events. For the three smaller instances, we report 25 deletions plus 25 additions; for the 100,000-customer case we also report 100 plus 100. The large-scale stress series varies the balanced batch from 50 to 12,800 total events. Reported algorithm time refers to the solver's recorded update phases and is distinguished from external wall time, which also includes surrounding application overhead. Only runs with completed route validation are reported as successful scale points. At 25,600 events, the ADD/repair audit recorded zero invalid routes, but refinement did not complete: the two-route shared-memory scratch request was 103,064 bytes, exceeding the 101,376-byte per-block limit recorded for that configuration. Larger stress tests likewise were not completed successful runs. Thus their omission reflects an execution/resource limit, not evidence that their computed routes failed the CVRP constraints.

7.3 Hyperparameters and experimental variants

Table 1 fixes the hyperparameters that otherwise make performance and quality hard to interpret. The same repair configuration is retained when refinement is changed from one to fifteen passes. All scheduler values describe targets or maxima; the actual blocks per launch depend on route count and device resources.

Setting	Value	Role
Routes per scheduler thread, ρ	8	Initial centroid-scan target
Local candidate budget, $h_{\max}$	5	Maximum cached route attempts
Route queue capacity	5	Pending requests per route
Refinement neighbors, K_r	5	Nearest route partners
Worker warp target	256	Maximum route workers
Scheduler packing target	8	Warps in a block, adapted to memory
Refinement budget, P	1 / 15	RUNI1 / RUNI15; early-stop enabled

Table 1: Runi parameters used in the reported comparisons.

7.4 Reference solvers and comparison scope

We consider two baselines that answer different questions. VND denotes Senay Patel's Dynamic-CVRP implementation [2], which performs an existing-solution dynamic update followed by a broader CPU variable-neighborhood descent. seqMDS is a static, independently optimized solution of the resulting final customer set. It is useful for observing how an independent restart compares with incremental repair, but its compute budget and timing boundaries are not identical to those of RUNI. The algorithms are therefore shown side-by-side rather than described as controlled hardware speedups. For final-state quality, we additionally consult BKS values for the exact full active instance, and distinguish them from an HGS rerun; an HGS solution on the starting subset is not a reference for the full final set. The HGS method follows Vidal [3, 4].

8. Results

We now examine how RUNI behaves as dynamic batches grow, how additional refinement passes affect cost and runtime, and how its results compare with the available baselines. We keep the mixed-update and full-instance reconstruction experiments separate because they answer different questions.

8.1 Scaling with the number of dynamic events

The first question is whether batch processing remains practical as the update itself grows. Figure 5 follows the same 500,000-customer base while the number of deletions and additions increases in equal proportions. The algorithm remains valid through 12,800 total events. A comparable original-instance static solve time for this 500,000-customer benchmark was not present in the supplied experiment logs; therefore no static baseline line can responsibly be added to Figure 5 without a separate matched run. Runtime increases approximately linearly over the successful observed points; a fixed refinement component and additional repair retries mean that the curve should not be interpreted as a strict complexity bound.

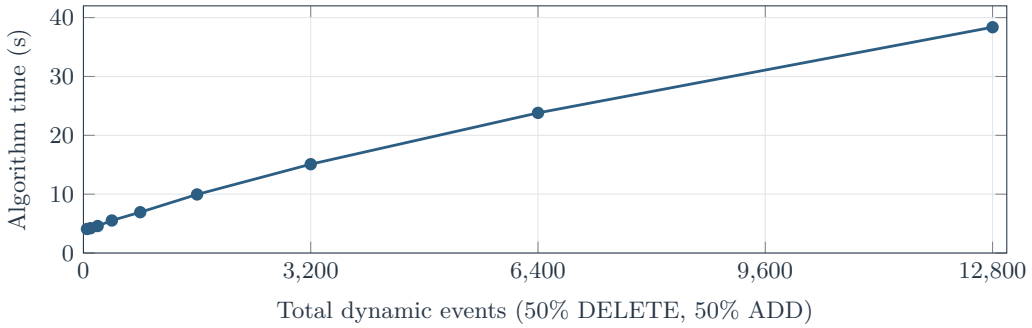


Figure 5: Measured RUNI1 batch-update scaling on XML500000_2366_05; every point represents a completed valid update; original-instance static runtime was not recorded.

The largest completed experiment contains 6,400 cancelled and 6,400 newly added customers, with 38.37 s of algorithm time. Beyond that point, refinement is bounded by its two-route temporary scratch allocation. In the 25,600-event attempt, the reserved stride reached 12,883 integers, giving $2 \times 4 \times 12{,}883 = 103{,}064$ bytes per warp for two temporary route sequences, beyond the recorded 101,376-byte shared-memory limit. The repair audit itself passed, but refinement could not launch in that configuration. These incomplete attempts are therefore not plotted as completed update timings. This is a tested implementation/resource limit, not a theoretical limit on

Events	ADDs	Algorithm (s)	Final cost
50	25	4.085	12,335,987
100	50	4.220	12,348,188
200	100	4.585	12,374,911
400	200	5.529	12,419,019
800	400	6.933	12,486,501
1,600	800	9.954	12,582,454
3,200	1,600	15.091	12,715,321
6,400	3,200	23.811	12,869,226
12,800	6,400	38.370	13,116,377

Table 2: Completed mixed-batch scaling results on XML500000_2366_05. All entries passed route validation; failed larger runs are excluded.

dynamic CVRP or on all possible GPU algorithms.

One controlled architectural comparison helps explain the scaling result. For the 12,800-event workload, changing a one-warp shared-memory block layout to eight-warp blocks with private global worker scratch reduced cooperating scheduler blocks from 88 to 11, while keeping 88 total scheduler warps. The corresponding ADD time fell from 206.32 s to 31.82 s; both runs passed post-repair and post-refinement route audits. The improvement arises from reducing synchronization across mostly waiting scheduler blocks, rather than from creating more nominal worker warps. These are recorded configurations, not averaged repetitions.

8.2 How many refinement passes are enough?

Increasing P expands the opportunity to improve a feasible repaired solution, but it also adds search time. Table 3 records final costs at four pass budgets on three different route-size regimes. The normalized progress curve in Figure 6 makes the diminishing returns visible without allowing the naturally larger absolute costs of the large instance to dominate the plot.

Instance	Passes	Final cost	Time (s)
X-n1001-k43	1	77,166	0.0305
	5	74,708	0.1353
	15	74,122	0.2734
	20	73,972	0.3136
XL-n10001-k1570	1	2,362,379	0.0438
	5	2,355,509	0.1453
	15	2,354,029	0.3951
	20	2,353,981	0.5200
XML100000_2222_01	1	12,827,656	2.506
	5	12,805,385	11.007
	15	12,799,397	31.779
	20	12,799,148	43.343

Table 3: Recorded refinement-pass sweeps: 25 ADD and 25 DELETE on the first two instances, and 100 ADD and 100 DELETE on the last. Separate from the 80/20 full-instance experiment.

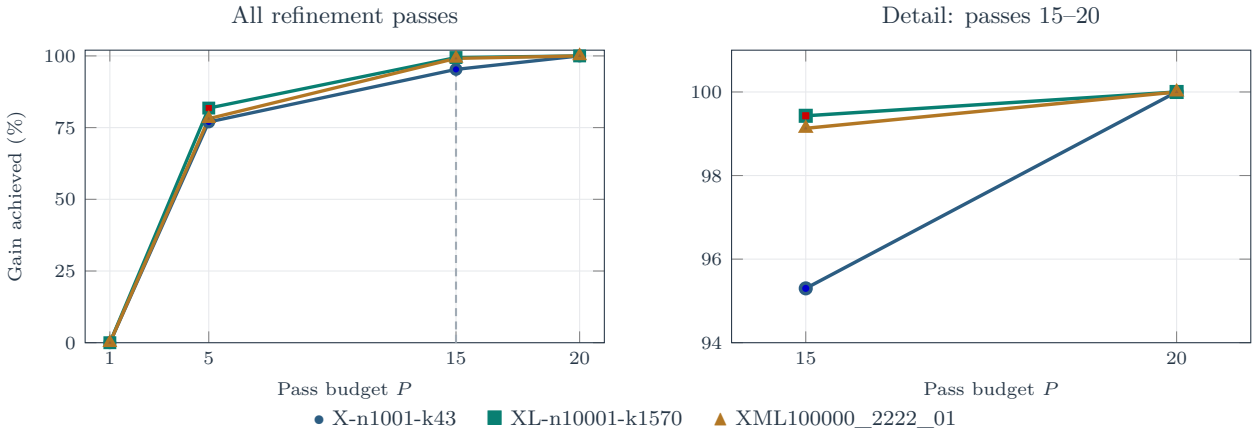


Figure 6: Share of the improvement achieved by $P = 20$ refinement passes. Left: all pass budgets. Right: a magnified view of the three results at $P = 15$ and their convergence at $P = 20$ (note the different vertical scales). At $P = 15$, the gains are 95.30%, 99.43%, and 99.13%, respectively.

By fifteen passes, the three sweeps have already achieved approximately 95.3%, 99.4%, and 99.1% of the cost reduction observed by twenty passes. The final five passes reduce the measured 100,000-customer workload's cost

by only 249 units while extending its recorded time from 31.78 to 43.34 s. This supports Runi15 as a practical saturation point for these observations, not an assertion that every workload has an optimum at exactly fifteen passes. Runi1 remains appropriate when update latency is the principal constraint.

8.3 Mixed-update comparison against VND and seqMDS

Table 4 and Figure 7 compare final costs for the same 25/25 workloads under four methods. Runi1 prioritizes update latency; Runi15 allows more refinement to recover solution quality. On the three 25/25 cases, Runi15 improves Runi1's cost and beats VND on two cases, while remaining close on the third. A fresh seqMDS restart reports higher costs and longer times in this record.

Instance and event batch	Method	Cost	Time (s)
X-n1001-k43 25/25	Runi1	77,428	0.02965
	Runi15	**74,122**	0.27340
	VND	75,014	6.82347
	seqMDS	80,277	8.33092
XL-n10001-k1570 25/25	Runi1	2,362,379	0.04818
	Runi15	**2,354,029**	0.39507
	VND	2,350,330	14.58895
	seqMDS	2,525,820	119.008
Flanders2 25/25	Runi1	4,564,957	2.84205
	Runi15	**4,468,369**	41.18131
	VND	4,502,981	81.04772
	seqMDS	4,910,320	387.659
XML100000_2222_01 100/100	Runi1	12,827,656	2.57730
	Runi15	**12,799,397**	31.779
	VND	—	—
	seqMDS	—	—

Table 4: Recorded mixed-update comparison (25 DELETE + 25 ADD; 100 + 100 for XML). Runi times are GPU update measurements; VND is a dynamic CPU baseline and seqMDS is a static restart. Times are not hardware- or search-budget-matched speedups. VND and seqMDS were not scalable to the XML 100/100 workload in the tested configurations, so their results are not reported there.

Relative to VND, Runi15's final cost is 1.19% lower on X-n1001-k43 and 0.77% lower on Flanders2; it is only 0.16% higher on XL-n10001-k1570. Its recorded update times are also shorter in all three cases (0.273 versus 6.823 s; 0.395 versus 14.589 s; 41.181 versus 81.048 s). Thus the fifteen-pass variant substantially narrows, and in two cases reverses, Runi1's quality disadvantage, at higher GPU update cost.

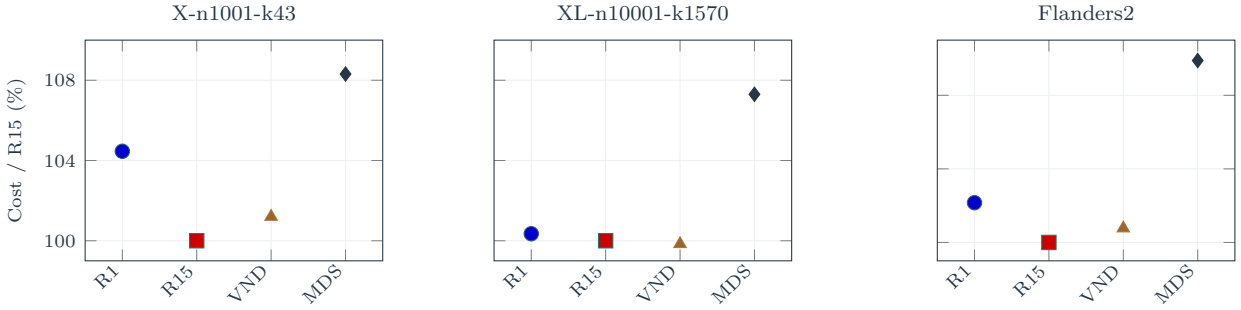


Figure 7: Final costs for the first three 25/25 mixed-update cases, normalized to Runi15 (R15 = 100%; lower is better). R1 = Runi1; MDS = seqMDS. All four methods use the same final customer set within each panel.

8.4 Quality relative to best-known full-instance solutions

A separate experiment starts with roughly 80% of a known CVRP customer set and adds the remaining 20%, so that the final active set reconstructs the full benchmark instance. Each 80/20 configuration was executed five times from the same starting solution and event set, using a refinement budget of $P = 15$. We report the mean final cost and sample standard deviation across the five runs, together with the minimum-to-maximum range of the five per-run gaps.

The official CVRPLib bound histories list the BKS values used in this comparison as 72,332 for X-n1001-k43 (updated 24 October 2026), 2,329,417 for XL-n10001-k1570 (updated 30 October 2026), and 4,364,450 for Flanders2 (updated 1 November 2026) [5]. The older reference routes stored in the benchmark archive are feasible reference solutions, not certified BKS values, and are not used as the denominator here. These percentages measure the distance from the current published BKS; they are not claims of gaps from proven optima.

Instance	Start / ADD	BKS	Runi15	SD(σ)	Gap range(%)
X-n1001-k43	800 / 200	72,332	77,901	553.4	7.14–8.99
XL-n10001-k1570	8,000 / 2,000	2,329,417	2,428,197	1,339.4	4.19–4.33
Flanders2	24,000 / 6,000	4,364,450	4,685,054	23,105.5	6.74–8.01

Table 5: Five-run 80/20 reconstruction results. Runi15 reports the mean final cost; SD is the sample standard deviation ($n-1$ denominator). Gap range is the minimum-to-maximum percentage above the current BKS across the five runs.

The mean gaps over the five runs are 7.70% for X-n1001-k43, 4.24% for XL-n10001-k1570, and 7.35% for Flanders2. The corresponding gap ranges show that the XL result is the most consistent in relative terms, while the X instance has the widest relative spread. The absolute standard deviations in Table 5 should be interpreted in the scale of each instance's objective value.

The average total GPU times across the five runs were 0.395 s, 1.291 s, and 28.457 s for X-n1001-k43, XL-n10001-k1570, and Flanders2, respectively. The corresponding mean external wall times were 0.618 s, 1.548 s, and 28.898 s. These are Runi15 timing measurements for the 80/20 workloads. The five-run dataset does not contain matching VND or seqMDS timings for these same 80/20 experiments, so an average speedup cannot be computed from this campaign. The 25/25 timings in Table 4 are a separate experiment and are not combined with these measurements.

A separate HGS Advance run [4] produced an initial route solution for the 800-customer subset of X-n1001-k43: cost 59,403, 34 routes, in 633.41 s. That number is useful for documenting the quality of the starting incumbent, but it cannot be placed into Table 5 as an HGS final cost, because the solver had not yet been shown to complete on the reconstructed 1,000-customer set under a matched final-state protocol. The appropriate HGS final column is therefore left unreported rather than inferred from the initial solution.

8.5 Diagnostic GPU profiling

Nsight Compute profiled the persistent `addWorkerPoolKernel` on two ADD workloads. For medium1, the archived profile showed approximately 0.18% compute SOL, 2.25% memory SOL, and 78.75% long-scoreboard stalls. For large5 with 200 ADDs, long-scoreboard stalls remained about 77.2%, achieved occupancy was 16.7% (50% theoretical), and eligible warps per scheduler were approximately 0.01. The larger workload therefore did not remove the main bottleneck: the kernel is limited more by memory-dependent waiting and lack of eligible warps than by peak arithmetic throughput.

Nsight Systems recorded a complementary end-to-end timeline for the same large5 ADD-200 run. Because a validated kernel-by-kernel time breakdown is not available here, this trace is not used to attribute a share of total runtime to the repair kernel. The Nsight Compute measurements nevertheless motivate a closer study of memory dependencies, polling and synchronization before increasing nominal arithmetic work.

9. Conclusion

- **Dynamic GPU-based route reoptimization.** Runi reuses incumbent routes for batched dynamic CVRP updates, separating deletion and capacity-feasible insertion/ejection from sparse GPU refinement. Its main implementation contributions are cached centroid candidate selection, a multi-block scheduler, route-exclusive worker warps dispatched through a ready-route queue, and matching-based local search. The same update machinery can also handle a demand change by replacing the old request with the new demand, checking route capacity, and performing either direct insertion or ejection when necessary. The recorded experiments demonstrate successful mixed batches through 12,800 events on a 500,000-customer instance, with fifteen refinement passes providing a practical quality–latency trade-off.

- **Runtime advantage over existing approaches.** In the three 25/25 mixed-update experiments, Runi15 consistently recorded lower execution times than both baselines. Relative to VND, the measured runtime ratios were 24.96×, 36.93×, and 1.97× for X-n1001-k43, XL-n10001-k1570, and Flanders2, respectively. Compared with seqMDS, the corresponding ratios were 30.47×, 301.23×, and 9.41×. These results demonstrate the potential latency advantage of incremental GPU-based reoptimization over CPU-based dynamic search and full static reoptimization.

- **Solution quality and five-run performance.** In the five-run 80/20 study, the mean Runi15 gaps to the current CVRPLib BKS values were 7.70%, 4.24%, and 7.35% for X-n1001-k43, XL-n10001-k1570, and Flanders2, respectively. Table 5 reports the corresponding sample standard deviations and per-run gap ranges. Mean external wall times were 0.618 s, 1.548 s, and 28.898 s, respectively, while mean total GPU times were 0.395 s, 1.291 s, and 28.457 s. These averages also quantify the observed run-to-run variation in solution quality.

10. Future Work

- **Extending Runi towards static optimization.** One promising extension is to construct a deliberately inexpensive, capacity-feasible but suboptimal initial solution (e.g., randomized grouping or a greedy sweep), and run Runi's refinement without requiring any customer events. This turns the *refinement stage* into a candidate static CVRP improvement engine, not a demonstrated best static solver. A fair experiment must account for initial construction plus refinement time and compare final costs against seqMDS, HGS, and BKS on identical full instances, across multiple seeds.

- **Reducing GPU memory latency and synchronization overhead.** The profiler results motivate further systems-level improvements. Nsight Compute reports substantial long-scoreboard stalls and very few eligible warps in the persistent repair kernel; future Nsight Systems analysis should quantify the resulting end-to-end overhead. Promising directions include reducing dependent memory accesses and queue polling, improving latency hiding, and making pair-refinement scratch allocation more resilient to long routes.

- **Improving reproducibility across GPU executions.** The five-run experiments quantify final-cost variation, but do not by themselves identify which stage creates it. Although the nearest-K route-pair list and matching rounds are ordered deterministically for a fixed route state, the ADD phase has multiple worker warps claiming queued work through atomic operations. Their completion order can vary, affecting which feasible request claims scarce route capacity first and hence the repaired incumbent seen by refinement. A different incumbent can produce different candidate pairs, accepted improvements, and the pass at which the zero-improvement early-exit condition is met. This is a plausible source of observed final-cost variation rather than evidence of an incorrect route or a confirmed data race. Future work should record route states immediately after repair and after each refinement pass, then run repeated experiments to isolate repair-stage and refinement-stage variability. Deterministic worker dispatch or canonicalized processing can then be evaluated when reproducibility is required.

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Reference Page: Terminology, Parameters, and Instance Sizes

This page collects the principal terms used in the method and the full benchmark identifiers. The table is a reading aid, not an additional experiment.

Glossary

Term	Meaning
Frontier	Device-resident requests waiting for insertion, including displaced customers.
Ready-route queue	Route-ID queue that tells workers exactly which routes have pending work.
Per-route priority queue	Bounded ordered requests for a single route.
Scheduler block	CUDA threads evaluating disjoint route candidates; block 0 also coordinates.
Worker warp	Thirty-two GPU lanes cooperatively and exclusively changing one route.
Route ownership	Atomic exclusion used during route mutation or queue insertion.
Route stride	Reserved slots per route, not actual route length.
Centroid cache/version	Spatial candidate list; version checks detect changes to cached routes.
LOCAL / DEPOT mode	Bounded cached-candidate exploration / exhaustive fallback.
Ejection	Displace a route customer and put that customer back on the frontier.
Matching round	Disjoint route pairs that are safe to refine concurrently.
BKS	Best-known full-instance route cost, not necessarily proven optimal.
RUNI1 / RUNI15	Same method with at most 1 / 15 refinement passes.

Refinement moves (VND neighborhoods)

The labels below refer to the move implementations used in RUNI. The VND baseline is Senay Patel's Dynamic-CVRP code [2]; the general VND framework is in [6], and SWAP* is the customer-exchange neighborhood of [3].

Move	Meaning
Move 8	Exchange suffixes of two routes, reversing the transferred tail orientation considered by this move.
Move 9	Exchange suffixes of two routes, retaining the corresponding tail orientation considered by this move.
Move 1	Relocate one customer from a route into a feasible insertion position of another route.
SWAP*	Exchange one customer from each of two routes, allowing each to be reinserted at its best feasible position rather than at the original positions [3].
Move 7 (2-opt)	Reverse a segment within one route to remove costly detours without changing its customer set.

Experimental hyperparameters (quick reference)

$\rho = 8$ routes/scheduler thread; $h_{\max} = 5$ cached route attempts; queue capacity $= 5$; $K_r = 5$ centroid-nearest refinement partners; worker cap $= 256$ warps; initial packing $= 8$ warps/block; $P = 1$ or 15 refinement passes, with early stop.

Instance register

Benchmark identifier	Customers	Q	Experimental role
X-n1001-k43	1,000	131	25/25; 80/20
XL-n10001-k1570	10,000	479	25/25; 80/20
Flanders2	30,000	200	25/25; 80/20
XML100000_2222_01	100,000	37	100/100 pass study
XML500000_2366_05	500,000	1,501	Scaling to 12,800 events

Customer count excludes the depot. In the underlying CVRPLIB files, the corresponding DIMENSION field includes one additional depot vertex. Values of Q were read from the instance headers.